Lab 2

Requirements Analysis of the TTPU Student Print Service

Student: Albert Muratbaev

Task 1 Requirements

ID	Requirement	F/N	How it would be tested
FR-01	The system shall give every new student 100 free pages at the beginning of each semester.	F	Register a student and verify a balance of 100 free pages.
FR-02	The system shall allow a student to submit a document as a print job.	F	Submit a valid document and verify that a job is created.
FR-03	The system shall reject a print job longer than 50 pages.	F	Submit 50-page and 51-page files; accept the first and reject the second.
FR-04	The system shall assign every accepted print job a unique job number.	F	Submit 100 jobs and verify that no job number repeats.
FR-05	The system shall deduct printed pages from the student's free-page balance before charging money.	F	Print 10 pages from a balance of 100 and verify a balance of 90.
FR-06	After the free balance reaches 0, the system shall charge 200 soum for each black-and-white page.	F	Print 3 black-and-white pages with zero balance and verify 600 soum.
FR-07	After the free balance reaches 0, the system shall charge 400 soum for each colour page.	F	Print 3 colour pages with zero balance and verify 1,200 soum.
FR-08	The system shall allow a student to cancel a job only while its status is Waiting.	F	Cancel one waiting and one printed job; only the waiting job is cancelled.
FR-09	The system shall show the operator all jobs whose status is Waiting.	F	Create waiting and completed jobs and verify only waiting jobs appear.
FR-10	The system shall let the operator mark a waiting job as Printed.	F	Mark a waiting job printed and verify its status and removal from the queue.
FR-11	At the end of each day, the system shall report the total number of pages printed.	F	Print known jobs and compare the report page total.
FR-12	At the end of each day, the system shall report the total money collected in soum.	F	Complete paid jobs and compare the report with calculated charges.
NFR-01	The operator shall complete submission-to-queue entry in no more than 3 screens and 5 clicks.	N Usability	Ask the operator to add 5 jobs and count screens and clicks.
NFR-02	The waiting-job list shall open within 2 seconds with 500 jobs stored.	N Performance	Load 500 jobs and measure the list response time.
NFR-03	Only an authenticated operator shall change a job status or view daily revenue.	N Security	Try both actions with student and operator accounts.
NFR-04	The system shall preserve all confirmed jobs and balances after an unexpected restart.	N Reliability	Create jobs, interrupt the server and verify all records after restart.
NFR-05	The operator application shall run in the two latest major versions of Chrome and Edge.	N Portability	Repeat the operator workflow in the specified browser versions.

Task 2 Requirement collection plan

Technique	From whom or where	What is still unknown
Interview	Department head, operator and accountant	Approval rules, payment handling, roles, exceptions and required reports.
Questionnaire	At least 100 students	File types, colour demand, expected volume and preferred status notifications.
Observation	Existing print room during busy periods	Actual service steps, bottlenecks, reprints, abandoned jobs and queue size.
Document analysis	Student lists, price rules, receipts and semester calendar	Identity fields, accounting evidence, semester dates and retention rules.
Prototyping	Operator and 5 students	Whether submission, queue and cancellation screens are understandable.

First choice: interview the operator. The operator performs the daily process and can quickly expose exceptions and practical constraints; observation should follow to verify that the described process matches reality.

Interview questions

Money

• How and when does a student pay for pages beyond the free allowance?

• Are unused free pages carried to the next semester?

• When is a refund or free reprint allowed?

Jobs and documents

• Which file formats and maximum file size must be accepted?

• May a job mix colour and black-and-white pages?

• What should happen to a paid job that is never collected?

Users and operation

• How is a student identified and who may correct a page balance?

• How should the queue work when the printer or network is unavailable?

Task 3 Repair another analyst's requirements

Item	Broken properties and reason	Corrected requirement	Test
a	Clear and testable: fast has no limit.	NFR-06 The waiting-job screen shall open within 2 seconds for 500 stored jobs.	Load 500 jobs and measure 20 openings.
b	Clear, complete and testable: handle is undefined and no actor or operation is named.	FR-13 The system shall allow an operator to find a student by university ID and view the current page balance.	Search a known ID and compare the displayed record.
c	Consistent: limits of 50 and 80 pages conflict.	FR-03 The system shall accept jobs of 1-50 pages and reject jobs of 51 pages or more.	Submit 1-, 50- and 51-page files.
d	Necessary and testable: modern and beautiful are subjective and not tied to a user need.	NFR-01 The operator shall add a job in no more than 5 clicks after signing in.	Perform 5 additions and count clicks.
e	Complete and testable: the pricing rules and inputs are absent.	FR-14 The system shall calculate 200 soum per black-and-white page and 400 soum per colour page after free pages are used.	Price a mixed known job and compare the total.
f	Consistent and complete: it conflicts with paid printing after the allowance.	FR-15 The first 100 pages per student per semester shall be free; later pages shall use FR-14 prices.	Use 100 pages, then print one additional page.
g	Clear, complete and testable: probably, some kind and something are ambiguous.	FR-16 The system shall show a cancellation confirmation containing the job number within 2 seconds.	Cancel 10 waiting jobs and verify message, number and time.
h	Necessary and traceable: it is an unsupported design decision, not a stakeholder requirement.	Do not rewrite as a requirement; move the database choice to the design document after architecture evaluation.	Review the design decision against volume, skills and maintenance constraints.

The two statements that should not be repaired merely by wording are (d) and (h). Statement (d) expresses subjective aesthetic taste with no established stakeholder need; first elicit a measurable usability need. Statement (h) prescribes implementation technology; evaluate it during design and record it in the design document only if justified.

Task 4 What the customer did not say

Missing situation	Requirement added
A student pays but never collects the output.	FR-17 The system shall change an uncollected printed job to Abandoned 3 working days after printing and include it in the daily report.
The printer jams after charging but before the full job prints.	FR-18 The operator shall mark a failed job for one free reprint of its unprinted pages.
A file is corrupt or has an unsupported format.	FR-19 The system shall reject an unreadable file before payment and show a reason.
A student submits the same job twice.	FR-20 The system shall warn when the same student submits a file with the same name and page count within 2 minutes.
The semester changes while balances still exist.	FR-21 An administrator shall reset every active student's free balance to 100 pages on the configured semester start date.
The operator makes an incorrect status or balance change.	FR-22 The system shall record operator ID, time, old value and new value for every manual status or balance change.

The most dangerous omission is charging for a job that fails during printing. If discovered only at deployment, students may be charged for pages they did not receive, causing disputes and loss of trust. The repair may require changes to requirements, transaction design, code, database records, accounting procedures and tests; using the lecture cost chart, a defect that costs about 1 unit in requirements can cost roughly 50-100 units after release.

Task 5 The project through six phases

Phase	What is done in this project	Output	Who does it
1 Requirement gathering	Interview the operator and department head, observe printing, survey students and define prices, queue, cancellation and reports.	Approved SRS with use cases and acceptance tests.	Analyst, customer, operator, students.
2 Feasibility study	Estimate six-week scope, reuse university accounts, check printer integration, legal handling of student data and operating cost.	Go/no-go report and limited version-1 scope.	Analyst, developer, IT administrator, department head.
3 Design	Design student submission, operator queue, pricing, balances, job states, database, security and reporting.	HLD, LLD, data model and interface prototype.	Developer/architect with operator review.
4 Coding	Implement authentication, uploads, page counting, queue, prices, cancellation, reporting and unit tests.	Source code, database migrations and unit tests.	One developer.
5 Testing	Test rules, prices, 50-page limit, 100-page balance, permissions, failures, performance and usability.	Test report, defects and acceptance result.	Developer, analyst, operator and pilot students.
6 Deployment and maintenance	Install in the print room, import users, train the operator, monitor, back up data and correct defects.	Working service, user guide, backups and maintenance log.	Developer, university IT and operator.

Feasibility answer: yes, but only for a small version 1 with the listed core functions and existing university infrastructure. The decision mainly depends on schedule feasibility because one programmer has only 6 weeks; it also depends on technical feasibility of document page counting and printer integration. Scope must be frozen, a simple web architecture reused and advanced features deferred.